

REPORT



Submitted by:

Sejal Bahri (Threat Detection & Classification Using Supervised Learning)
Avishi Singla (Phishing detection)

Submitted to:

Ms. Neeraj Sharma

**Centre for Development of Advanced Computing A-34, Industrial Area,
Phase-VIII Sahibzada Ajit Singh Nagar (Mohali), Punjab – 160071)**

Threat Detection & Classification Using Supervised Learning

What is it?

Supervised learning trains models on labelled datasets of malicious and benign files to automatically detect and classify cyber threats in real time, faster and at greater scale than human analysts.

How it Works for Malware Detection

- **Data Collection:** Malware samples (VirusTotal, MalwareBazaar) and clean files are collected and labelled.
- **Feature Extraction:** Static features (file size, PE headers, APIs) and dynamic features (system calls, network activity) are pulled from each file.
- **Model Training:** Algorithms learn to separate malicious from benign patterns using these features.
- **Classification:** At runtime, the model scores a new file and triggers an alert if it crosses the threat threshold.
- **Retraining:** Models are regularly updated with new samples to prevent model drift.

Key Algorithms Used

Algorithm	Use Case
Random Forest	Endpoint protection (CrowdStrike)
SVM	Network intrusion detection
XGBoost	SIEM-integrated threat scoring
CNN	Malware binary image classification
LSTM	Sequential system call analysis

Real-World Case Studies

- **Microsoft Defender ATP:** Detected WannaCry variants 24 hours before signature-based tools using models trained on billions of Windows telemetry files. (*Microsoft Security Intelligence Report, Vol. 24, 2019*)
- **Cylance PROTECT:** Replaced signatures with a Random Forest model trained on 1B+ malware attributes, achieving 99%+ pre-execution detection. (*Cylance Threat Report, 2018*)
- **Darktrace NDR:** Identified APT lateral movement in network traffic using labelled flow data, deployed across 6,000+ organisations. (*Darktrace Annual Threat Report, 2023*)

Challenges

- **Adversarial attacks** — malware crafted to evade classifiers
- **Label quality** — mislabelled data reduces accuracy
- **Class imbalance** — benign files far outnumber malware samples
- **Model drift** — accuracy degrades as malware evolves

Phishing Detection:

Phishing detection uses supervised learning to identify fake emails, websites, messages, and login pages by training models on labeled examples of legitimate and malicious content.

In 2026, this area is becoming more important because AI-generated images, deepfakes, and realistic fake websites make phishing harder for humans and traditional filters to detect.

AI-generated images can be used to copy company logos, login screens, invoices, support chats, and brand visuals, which makes scams look trustworthy.

Because of this, phishing detection is increasing in importance, not decreasing.

Why AI-Generated Images Matter?

Attackers can use AI-generated images to create fake brand identities and convincing phishing pages.

Traditional detection methods that only check text or URLs are not enough anymore because the page may look visually real even if the code is suspicious.

Visual deception is now a major part of phishing, especially in fake login pages and impersonation attacks.

Companies Working on This:

Microsoft is using AI-based security systems to detect phishing, business email compromise, and attacker-in-the-middle campaigns across email, identity, and cloud services.

Check Point is using AI-powered phishing detection that analyzes many features in real time to block phishing sites and zero-day threats.

Graphus / INKY provides email security with AI-based phishing protection and computer-vision tools to detect spoofed brands and design-based impersonation.

Security research teams and academic papers in 2026 are also working on screenshot-based and image-based phishing detection using deep learning models.

What We Can Do in 2026?

- Use multilayered detection, not just one method.
- Check emails, URLs, webpage screenshots, logos, and layout patterns together.
- Use MFA or passkeys so stolen passwords alone are not enough.
- Monitor suspicious login behavior, inbox rule changes, and unusual forwarding activity.
- Train users to verify urgent payment requests, login pages, and brand-related messages carefully.
- Combine AI detection with human awareness because attackers now use polished, realistic content that is harder to spot.

Conclusion:

Phishing detection is more important in 2026 because AI-generated images and deepfakes make scams look real.

Companies are responding with AI-based, visual, and behavior-based security tools.

The best defense is a combination of supervised learning, visual analysis, identity protection, and user awareness.